



Chapter 1 and 2

CS 101- Programming
Fundamentals

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Know your instructor

- Neelma Bhatti, Assistant prof at DSSE
- Fresh PhD, Fresh fracture.
- New to Habib, (somewhat) new to Karachi.
- Mother to (an almost) 5-year-old.

Know your course

- **CS 101 – PFun** : Equip you with problem solving skills using Python as a language tool.
- [Syllabus](#): Answers one-third of your course related queries.
- [Textbook](#): Has answers to another one-third of them.
- **Instructor/ TAs/ RA**: Happy to answer the remaining one-third of the queries.

Recap: Chapter 1

What is a program?

- sequence of instructions that specifies how to perform a computation.
- **Input, output, math, conditional execution, repetition**
- Running Python: Interpreter
- Arithmetic operators
- Values (letters,numbers) and types (int,float,string,bool)

Recap: Chapter 1

- Natural (English, urdu, french, etc.) and formal (discrete maths, Java, Python etc.) languages:
 - *Programming languages are formal languages that have been designed to express computations.*
 - Syntax rules
 - ambiguity, redundancy, literalness
- Debugging: Finding and eliminating bugs/errors

Chapter 2: Variables, expressions and statements

- **Variable:** *A name referring to a value.*
- **Assignment statements:** *creates a new variable and gives it a value*

```
message → 'And now for something completely different'  
n → 17  
pi → 3.1415926535897932
```

- Variable names: Meaningful, legal names (should not: be Python keywords, begin with numbers, contain illegal characters)

Chapter 2: Variables, expressions and statements

- **Expressions:** combination of values, variables, and operators
 - Is evaluated
- **Statements:** unit of code that has an effect (creating a variable, displaying a value)
 - Is executed

```
>>> 42
42
>>> n
17
>>> n + 25
42
```

```
>>> n = 17
>>> print(n)
```

Chapter 2: Variables, expressions and statements

- **Order of operations** follows PEMDAS, from left to right
- **String operations:** Concatenation (+) and repetition (*)
- **Comments:**
 - #document non-obvious features of the code
 - #Ignored by interpreter
- **Debugging:** Finding and eliminating *bugs*
 - Syntax, runtime, semantic errors